The Random Forest Algorithm

Ensemble Learning with Decision Trees

Why Random Forest?

A **decision tree** is a simple model that learns to ask yes/no questions about data, like a flowchart. We treat it as a **black box**: feed it data, it automatically learns question rules, and outputs predictions. However, a single tree suffers from **high variance** — small changes in training data produce wildly different trees.

Random Forest applies the **wisdom of crowds**: just as averaging many people's guesses beats any single guess, combining many *diverse* decision trees produces much better predictions than any single tree.

Definition: Random Forest

A **Random Forest** is an **ensemble method** that combines multiple decision trees:

- Builds k decision trees, each trained on different data via **sampling with replacement** (formally called "bootstrap sampling")
- At each split, considers only m randomly selected features (not all of them), creating diverse trees with different "perspectives"
- Aggregates predictions by **majority vote** (classification) or **averaging** (regression)

Two sources of randomness — different *data samples* and different *feature subsets* — ensure the trees are diverse.

Random Forest Algorithm

1. **For** each tree $i = 1, \ldots, k$:
   (a) **Sample data**: Draw n items with replacement from training data (some items appear twice, some are left out)
   (b) **Build tree**: At each node, randomly select m features; split on the best among them
   (c) **Grow fully**: Expand the tree to maximum depth (no pruning)
2. **To predict**: Pass input $\mathbf{x}$ through all k trees, return majority vote: $\hat{y} = \text{mode}\{h_1(\mathbf{x}),\ h_2(\mathbf{x}),\ \ldots,\ h_k(\mathbf{x})\}$

Sampling with Replacement

Each tree trains on n samples drawn *with replacement*. This means some data points appear multiple times, while others are left out entirely. On average, $\sim 63.2\%$ of unique training points are included.

Probability of being left out:

$$P(\text{not picked}) = \left(1 - \frac{1}{n}\right)^n \approx 0.368$$

Free Validation: Out-of-Bag Samples

The data points that a tree *never saw* during training ($\sim 36.8\%$) are called **out-of-bag (OOB)** samples. We test each tree on its own unseen samples to get a **free accuracy estimate** — no separate test set needed!

This is similar in accuracy to k-fold cross-validation.

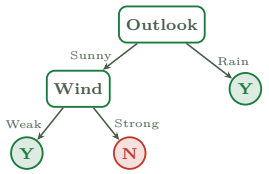
Step-by-Step Example

Using a hiking dataset with features: **Outlook**, **Temperature**, **Humidity**, **Wind** → **Go Hiking?**

We build a tiny forest: 3 trees, each split considers 2 of 4 features.

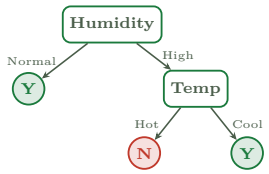
Tree 1 Outlook & Wind

Sampled rows: {1,1,3,4,5,7,7,8}
Random features at root: {Outlook, Wind}



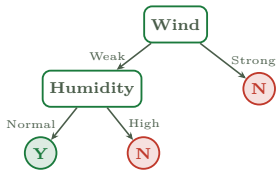
Tree 2 Humidity & Temp

Sampled rows: {2,3,3,5,6,6,8,9}
Random features at root: {Humidity, Temperature}



Tree 3 Wind & Humidity

Sampled rows: {1,2,4,4,6,7,9,9}
Random features at root: {Wind, Humidity}



Aggregation: Majority Vote

New data point:
Outlook=Sunny, Temp=Cool, Humidity=Normal, Wind=Weak

Tree	Path through tree	Vote
Tree 1	Outlook=Sunny → Wind=Weak	**Yes**
Tree 2	Humidity=Normal	**Yes**
Tree 3	Wind=Weak → Humidity=Normal	**Yes**

Final vote: 3–0 in favor of Yes.
The Random Forest predicts: **Yes — go hiking!**
Even if one tree had disagreed, majority vote would smooth out the error.

Feature Importance

How do we know which features matter most? Simple test:

1. Record the forest's accuracy on unseen (out-of-bag) data
2. For one feature: randomly **shuffle** its values, breaking its relationship with the outcome
3. Re-evaluate accuracy — if it drops a lot, that feature was important!
4. Repeat for each feature and rank by drop size

This is called **permutation importance**. The bigger the accuracy drop, the more the model depends on that feature.

Key Takeaways

1. **Diversity reduces variance**: Sampling with replacement + random feature selection create diverse trees whose errors cancel out
2. **Free validation**: Out-of-bag samples provide a free accuracy estimate — no separate test set needed
3. **Interpretable**: Permutation importance reveals which features drive the model's predictions